IGCSE 0607 Practice Questions — E5.3 — Combined Questions & Markscheme

Similarity

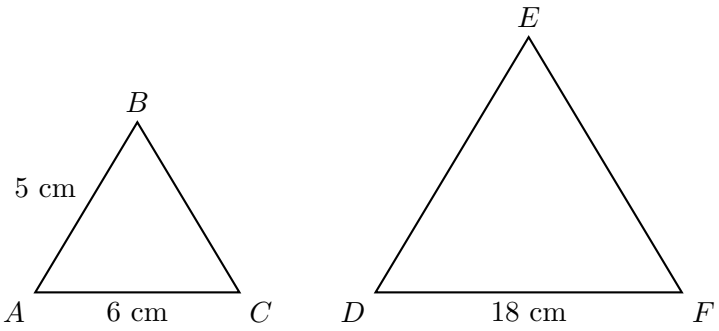
Total Questions: 6 — **Total Marks:** 42 Marks

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- **E5.3** Similarity
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*

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1 Triangle ABC and triangle DEF are mathematically similar.



NOT TO SCALE

(a) Calculate the length DE.

$DE =$ ..... cm **[2]**

(b) The area of triangle DEF is 72 cm^2.
Calculate the area of triangle ABC.

..... cm^2 **[2]**

2 Two mathematically similar solid cylinders are shown as Cylinder A and Cylinder B.
Cylinder A has a height of 6 cm.
Cylinder B has a height of 15 cm.

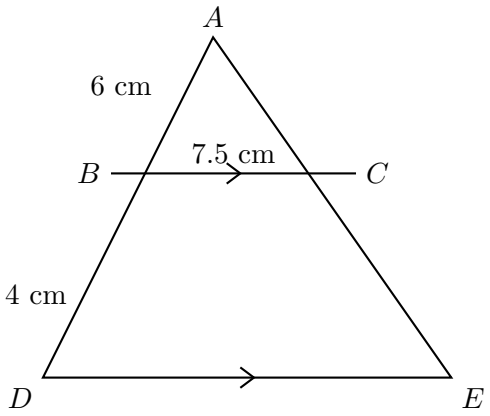
(a) The volume of Cylinder A is 48 cm^3.
Calculate the volume of Cylinder B.

..... cm^3 **[2]**

(b) The curved surface area of Cylinder B is 200 cm^2.
Calculate the curved surface area of Cylinder A.

..... cm^2 **[2]**

3 The diagram shows triangle ADE with a line BC drawn parallel to DE.
$AB = 6$ cm, $BD = 4$ cm and $BC = 7.5$ cm.



NOT TO SCALE

(a) Explain why triangle ABC is mathematically similar to triangle ADE.

[2]

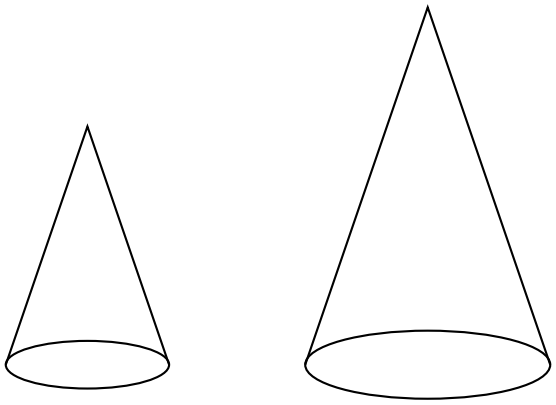
(b) Calculate the length of DE.

$DE =$ ..... cm **[2]**

(c) The area of triangle ABC is 18 cm^2.
Calculate the area of the quadrilateral $BCED$.

..... cm^2 **[3]**

4 Cone A and Cone B are mathematically similar solids.



NOT TO SCALE

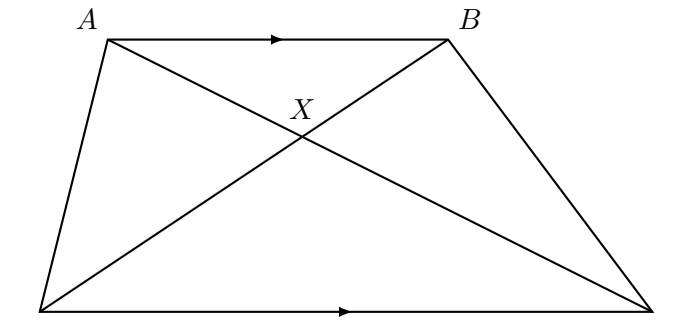
(a) The volume of Cone A is 108 cm^3 and the volume of Cone B is 500 cm^3.
The sum of the surface areas of Cone A and Cone B is 340 cm^2.
Calculate the surface area of Cone B.

..... cm^2 **[4]**

(b) Cone B has a total vertical height of 20 cm.
A smaller cone, mathematically similar to Cone B, is removed from the top of Cone B to leave a frustum.
The volume of the frustum is $\frac{7}{8}$ of the volume of Cone B.
Calculate the vertical height of the removed small cone.

..... cm **[3]**

5 The diagram shows a trapezium $ABCD$ with AB parallel to DC.
The diagonals AC and BD intersect at point X.
$AB = 9$ cm, $DC = 15$ cm and $BX = 6$ cm.
The area of triangle ABX is 27 cm^2.



NOT TO SCALE

(a) Explain why triangle ABX is similar to triangle CDX.

[2]

(b) Calculate the length of BD.

..... cm **[3]**

(c) Calculate the area of triangle CDX.

..... cm^2 **[2]**

(d) Calculate the area of triangle BCX.

..... cm^2 **[2]**

6 Solid Cone A has a base radius of 10 cm and a perpendicular height of 24 cm.

(a) Calculate the total surface area of Cone A.
Give your answer in terms of π.

..... cm^2 **[3]**

(b) Cone B is mathematically similar to Cone A.
The volume of Cone B is 100π cm^3.
Calculate the total surface area of Cone B.
Give your answer in terms of π.

..... cm^2 **[4]**

(c) A frustum is made by cutting a smaller cone, which is geometrically identical to Cone B, from the top of Cone A.
This frustum is melted down and recast completely into a solid sphere.
Calculate the radius of this sphere.

..... cm **[4]**

1 **(a)** Scale factor $= \frac{18}{6} = 3$ **[M1]**

15 **[A1]**

(b) Area scale factor $= 3^2 = 9$ **[M1]**

$\frac{72}{9} = 8$ **[A1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** Volume scale factor $= \left(\frac{15}{6}\right)^3 = \frac{125}{8}$ **[M1]**

$V_B = 48 \times \frac{125}{8} = 750$ **[A1]**

(b) Area scale factor $= \left(\frac{15}{6}\right)^2 = \frac{25}{4}$ **[M1]**

$A_A = 200 \div \frac{25}{4} = 32$ **[A1]**

[Back to Question] *[Total Marks: 4]*

3 **(a)** Angle A is common to both triangles **[B1]**

Angle ABC = Angle ADE (corresponding angles as BC is parallel to DE) **[B1]**

(b) $\frac{DE}{7.5} = \frac{6+4}{6}$ **[M1]**

12.5 **[A1]**

(c) Area scale factor $= \left(\frac{10}{6}\right)^2 = \frac{25}{9}$ **[M1]**

Area of $ADE = 18 \times \frac{25}{9} = 50$ **[M1]**

Area of $BCED = 50 - 18 = 32$ **[A1]**

[Back to Question] *[Total Marks: 7]*

4 **(a)** Volume ratio $A : B = \frac{108}{500} = \frac{27}{125}$ **[M1]**

Area ratio $= \left(\sqrt[3]{\frac{27}{125}}\right)^2 = \left(\frac{3}{5}\right)^2 = \frac{9}{25}$ **[M1]**

Let surface areas be $9x$ and $25x$. $9x + 25x = 340 \implies 34x = 340 \implies x = 10$ **[M1]**

$25 \times 10 = 250$ **[A1]**

(b) Volume of small cone $= 1 - \frac{7}{8} = \frac{1}{8}$ of Cone B **[M1]**

Ratio of heights $= \sqrt[3]{\frac{1}{8}} = \frac{1}{2}$ **[M1]**

Height of small cone $= 20 \times \frac{1}{2} = 10$ **[A1]**

[Back to Question] *[Total Marks: 7]*

5 **(a)** Angle AXB = Angle CXD (vertically opposite) **[B1]**

Angle ABX = Angle CDX (alternate angles, AB parallel to DC) **[B1]**

(Accept Angle BAX = Angle DCX for the second B1 mark)

(b) Length scale factor $= \frac{15}{9} = \frac{5}{3}$ **[M1]**

$DX = 6 \times \frac{5}{3} = 10$ **[M1]**

$BD = 6 + 10 = 16$ **[A1]**

(c) Area scale factor $= \left(\frac{15}{9}\right)^2 = \frac{25}{9}$ **[M1]**

Area $= 27 \times \frac{25}{9} = 75$ **[A1]**

(d) Ratio of bases $\frac{CX}{AX} = \frac{15}{9} = \frac{5}{3}$ **[M1]**

Area $= 27 \times \frac{5}{3} = 45$ **[A1]**

[Back to Question] *[Total Marks: 9]*

6 **(a)** Slant height $l = \sqrt{10^2 + 24^2} = 26$ **[M1]**

Surface Area $= \pi(10)^2 + \pi(10)(26)$ **[M1]**

360π **[A1]**

(b) $V_A = \frac{1}{3}\pi(10)^2(24) = 800\pi$ **[M1]**

Volume Ratio $= \frac{100\pi}{800\pi} = \frac{1}{8}$ **[M1]**

Area Ratio $= \left(\frac{1}{8}\right)^{\frac{2}{3}} = \frac{1}{4}$ **[M1]**

Surface Area of $B = 360\pi \times \frac{1}{4} = 90\pi$ **[A1]**

(c) Volume of frustum $= 800\pi - 100\pi = 700\pi$ **[M1]**

$\frac{4}{3}\pi R^3 = 700\pi$ **[M1]**

$R^3 = 700 \times \frac{3}{4} = 525$ **[M1]**

$R = \sqrt[3]{525} = 8.07$ (accept $8.066\ldots$) **[A1]**

[Back to Question] *[Total Marks: 11]*